5. *(a)* Gwyn carried out an experiment to find the solubility of potassium nitrate at room temperature, 25 °C. The method he used is shown in stages **1**-**3**.

Stage **1**

He added solid potassium nitrate to 50 cm^3 of water until a saturated solution was formed.

⇩

Stage **2**

He pipetted 25 cm^3 of the saturated solution into a pre-weighed evaporating basin. He put the basin and the solution into a warm oven.

⇩

Stage **3**

He removed the basin from the oven and allowed it to cool. He weighed the basin and crystals after 2 hours.

His results are shown below.

Mass of evaporating basin	= 42.6 g
Mass of evaporating basin and potassium nitrate crystals	= 54.2 g

(i) State what is meant by a *saturated* solution. [1]

....

....

(ii) Use Gwyn's results to calculate the solubility of potassium nitrate in g / 100 g of water. [2]

solubility = ... g / 100 g of water

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(iii) Gwyn's experimental value for the solubility of potassium nitrate is much bigger than expected. Suggest how Gwyn should adapt stage **3** of his method in order to get a more accurate value. Explain your reasoning. [3]

....

....

....

....

....

(iv) Gwyn was asked to find out if sodium chloride is more soluble in water than potassium nitrate. State which variable Gwyn **must keep the same** in his experiment in order for him to be able to compare the two solubilities. [1]

....

(b) (i) The solubility of potassium bromide at 20 °C is 64 g / 100 g of water. Calculate the mass of solid that forms when a solution containing 43.9 g of potassium bromide in 50 g of water is cooled to 20 °C. [2]

mass = ... g

(ii) Calculate the number of moles of potassium bromide, KBr, in 64 g. Give your answer to **two** significant figures. [2]

number of moles = ... mol

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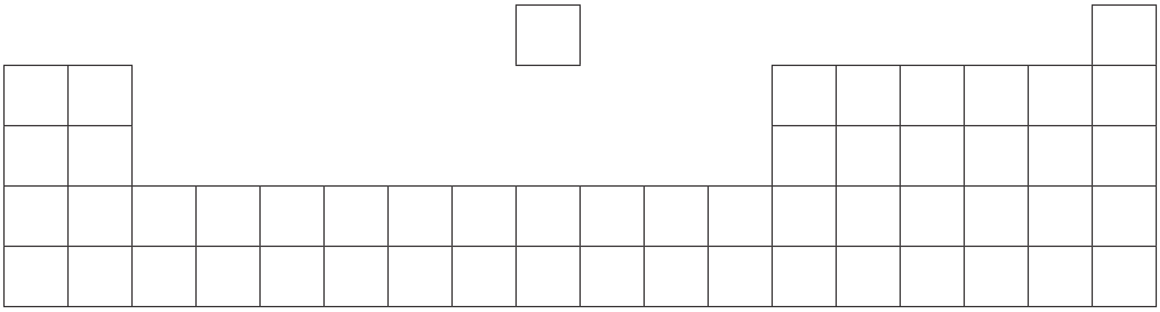
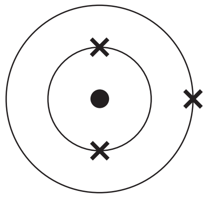
5. The following diagram is an outline of the Periodic Table.



(a) Write letters **A**, **B** and **C** on the **diagram** in the positions of the elements that fit the following descriptions. [3]

A the element with the electronic structure



B the element in Group 2 and Period 4

C an element that shows both metallic and non-metallic properties

(b) The following table shows the atomic numbers and boiling points of the inert gases.

Inert gas	helium	neon	argon	krypton	xenon
Atomic number	2	10	18	36	54
Boiling point (°C)	−269	−246	−186	−153	−108

(i) Plot this data on the grid opposite. Draw a suitable line. [3]

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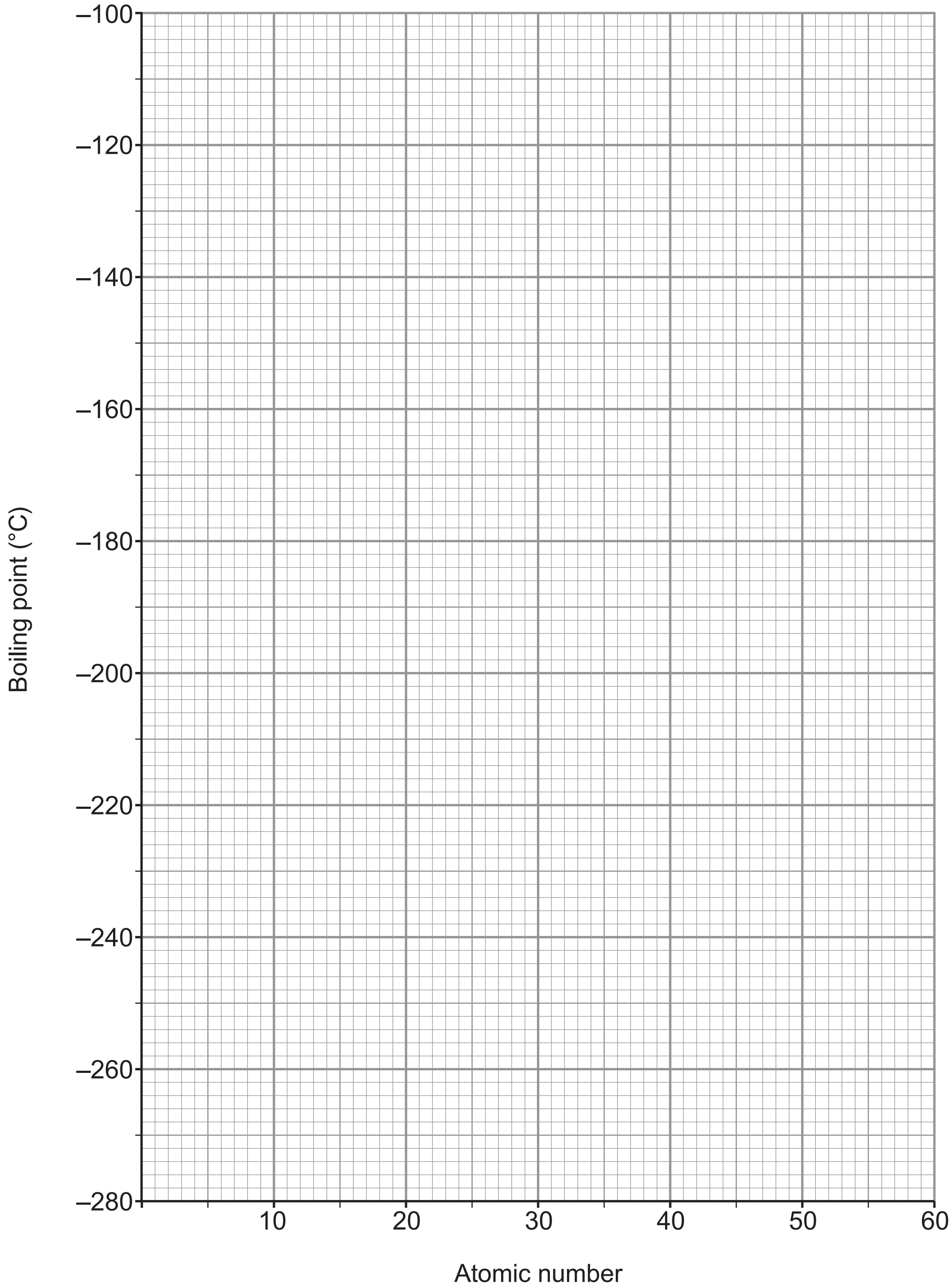

(ii) Describe the trend in boiling point shown on the graph. [1]

....

....

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(c) The following table shows the boiling points of the inert gases in both °C and K.

Inert gas	helium	neon	argon	krypton	xenon
Boiling point (°C)	−269	−246	−186	−153	−108
Boiling point (K)	4	27	....	120	165

Use the information in the table to calculate the boiling point of argon in K. [2]

Boiling point = ... K

(d) Give **one** use of argon. Explain in terms of electronic structure why it is used for this purpose. [2]

....

....

....

11

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5. The following table shows the solubility of potassium permanganate in water at temperatures between 30 °C and 60 °C.

Temperature (°C)	Solubility (g / 100 g of water)
30	9.0
35	10.8
40	12.5
45	14.4
50	16.8
55	19.2
60	22.2

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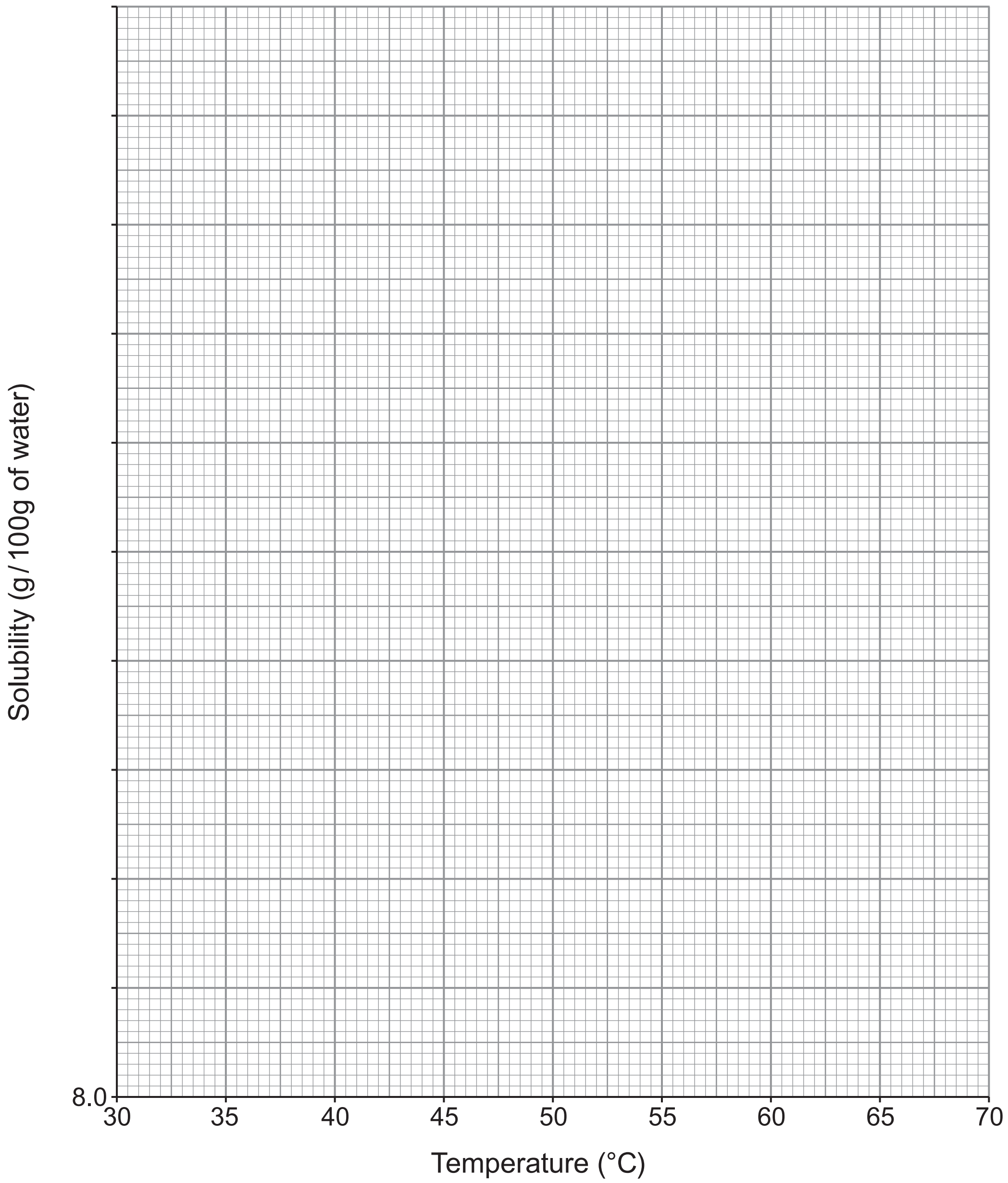
(a) Plot these values on the grid below. Draw a suitable line. [4]



(b) Use the graph to calculate the mass of crystals formed when a saturated solution in 500 g of water is cooled from 65 °C to 30 °C. [3]

Mass of crystals = ... g

7

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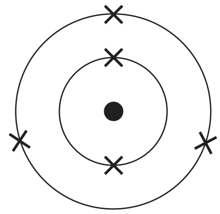
5. (a) The diagram shows an atom of element **X**.



Draw the electronic structure of the element directly **below** element **X** in the Periodic Table. [1]

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(b) The table below shows information about particles **A-F**.

The letters **A-F** are **not** the chemical symbols of the particles.

Particle	Number of protons	Number of electrons	Number of neutrons
A	9	10	10
B	6	6	8
C	7	7	7
D	6	6	6
E	9	9	10
F	3	2	4

(i) Give the **letters** of **two** particles which are **isotopes** of the same element. Explain your answer. [2]

Letters .... and ....

....

....

(ii) Give the **letters** of **two** particles which are **ions**. Explain your answer. [2]

Letters .... and ....

....

....

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5

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5. Temporary hard water is caused by the presence of dissolved calcium hydrogencarbonate, $Ca(HCO_3)_2$.

The table shows three different methods of removing temporary hardness from water.

Method	Product(s) in softened water
1. Adding sodium carbonate	insoluble calcium carbonate dissolved sodium hydrogencarbonate
2. Boiling	insoluble calcium carbonate carbon dioxide
3. Passing through an ion exchange column containing Na^+ ions	dissolved sodium hydrogencarbonate

(a) Complete the symbol equation for the reaction taking place in Method **1**. [1]

$Ca(HCO_3)_2$ + ... $\longrightarrow$ $CaCO_3$ + $2NaHCO_3$

(b) Underline the ratio of calcium ions to sodium ions exchanged in Method **3**. [1]

2:1 **1:1** **1:2** **2:2**

(c) Give the number of the method which does **not** form limescale. Give the reason for your answer. [2]

Number ....

Reason ....

....

(d) Tick (✓) the box next to the name of a compound that causes permanent hardness when dissolved in water. [1]

calcium sulfate ☐

potassium sulfate ☐

magnesium hydrogencarbonate ☐

sodium sulfate ☐

5

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